

KITEWRIGHT RANGE · REQUIREMENTS

Kitewright Range requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-range		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-04	Amish Chadha	Status from KWR-CAL-001 v0.4 after Amish's round-3 decision 10A (KWR-DDR-004; Amish: "For round 3, I agree with all your proposed recommendations"): the 0.99 kg Kitewright Core to the family envelope; targets unchanged
0.4	2026-10-03	Amish Chadha	Status from KWR-CAL-001 v0.3 after Amish's 2026-10-03 decisions (KWR-DDR-003; Amish: "i agree with all the 46 recommendations you provided. please proceed."); targets unchanged; R2 and R7 shortfalls accepted for the first prototype
0.3	2026-10-03	Amish Chadha	TRL 3 status from KWR-CAL-001 v0.2 on the constructable design; R10 cost stated against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2 concept status for every requirement
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Range requirements

Eight of the ten requirements are met on paper or by design (KWR-CAL-001 v0.4, Table 6). On 2026-10-04 Amish decided the Kitewright family reconciliation as recommended ("For round 3, I agree with all your proposed recommendations"; KWR-DDR-004): Range now carries the 0.99 kg Kitewright Core to the family mounting envelope, under the fuselage floor at the centre of gravity, with AS150 plugs on its packs. No requirement is restated. Endurance (R2) and take-off mass (R7) are not met. The targets below are unchanged. Amish decided the shortfalls on 2026-10-03 ("i agree with all the 46 recommendations you provided. please proceed."; KWR-DDR-003): 22 in propellers on 5215-class motors for R1, 5.0 Ah cells in the same two packs for R2 and R3, and the foam, glass and lite-ply airframe kept for the first prototype and weighed at TRL 4 for R7. No requirement is restated.

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (KWR-CAL-001)
R1	Vertical take-off and landing at high altitude	Hover with at least 30% thrust margin at 0.74 kg/m ³ air density (5,000 m) at maximum take-off mass	Lift motor and propeller thrust measured in AltiRig at 5,000 m density; margin calculated for the as-built mass	Met on paper: 48.3 % on maker-class thrust at 11.35 kg (22 in propellers, 5215-class motors); to measure in AltiRig
R2	Cruise endurance at altitude	At least 45 min wing-borne flight at 5,000 m and -20 C with a 1 kg payload and 20% reserve (target)	Energy model checked against bench power data, then timed flight at a high site	Not met: 30.6 min with 5.0 Ah cells and the 0.99 kg Core under the floor; accepted for the first prototype (KWR-DDR-003)
R3	Survey range	At least 40 km of survey track per flight at 5,000 m (target)	Flight log distance on a mapped survey pattern	Met on paper: 41.5 km
R4	Payload capacity	1.0 kg on the Kitewright mount within the centre of gravity range	Weigh and balance check; flight with a dummy payload	Met by design: the Core and its payload shoe under the CG, 28.3 % of the mean chord with or without the payload
R5	Operating temperature	-20 to +40 C (-4 to +104 F) for all airframe electronics and servos	Cold chamber soak and function test of servos, motors and avionics	Met on paper by specification; cold ratings to confirm when parts are bought
R6	Wind tolerance	Take off, transition and land in sustained wind of 10 m/s with gusts to 14 m/s	Flight tests with a logged anemometer at the launch site	Met on paper: cruise 22.6 m/s, stall 18.1 m/s, about 9 degrees of tilt holds a 14 m/s side gust in hover
R7	Maximum take-off mass	No more than 8 kg including payload (target)	Weigh the complete aircraft	Not met: 11.35 kg; airframe kept and weighed at TRL 4 (KWR-DDR-003)
R8	Transport	Packs into cases of no more than 1.3 m long; each carried load no more than 15 kg	Pack and measure; two-person carry trial	Met by design: longest part 1,205 mm; two cases of about 9 and 12.5 kg
R9	Patent design-arounds held	Fixed motors, single wing, rotor discs clear of the wing-top plane in every revision	Design review checklist at each revision	Met by design: discs 77 mm below the wing-top plane, checked by the model
R10	Open and buildable	All parts off the shelf or made with hobby tools; bill of materials at or below USD 5,000	Costed BOM review; independent build by a second team	Met on paper: Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 4,314 (USD 686 under the target)

■ Requirements at risk or not met

- R2 and R7 pull against each other: more pack energy adds mass, and a lighter aircraft carries less energy. Amish accepted both shortfalls for the first prototype (KWR-DDR-003); with the reconciled Core no option inside the

concept meets R2 on paper any more; a moulded carbon airframe with two 6S4P packs comes closest (42.5 min, USD 5,754) and is the second-prototype path (KWR-CAL-001 section H).

- R1 still rests on maker-class thrust that has not been measured at 5,000 m density; the 48.3 % margin leaves room for a thrust shortfall of about 12 % before the 30 % line.

■ Assumptions

- Hover and cruise figures are at the ISA density for 5,000 m (0.736 kg/m³) with the ColdCell packs warmed before take-off.
- Stock PX4 quadplane firmware is enough; no custom flight control code is needed (KWR-DDR-001, D1).
- ColdCell packs deliver 95 % of their capacity at -20 °C once warmed (cross-repo assumption).
- Most survey missions can start from a site within 10 km of the target lake.